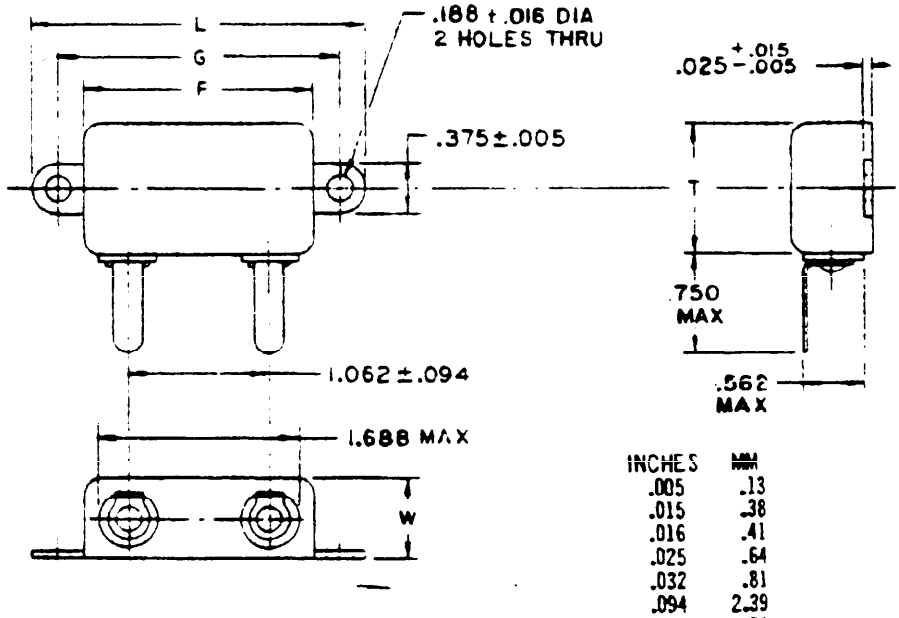
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, METALLIZED (PAPER, PAPER PLASTIC, OR PLASTIC FILM) DIELECTRIC, DIRECT CURRENT, (HERMETICALLY SEALED IN METAL CASES), STYLE CH53

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-18312.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

INACTIVE FOR NEW DESIGN AFTER 8 January 1970.



INCHES	MM
.005	.13
.015	.38
.016	.41
.025	.64
.032	.81
.094	2.39
.188	4.78
.375	9.53
.562	14.27
.750	19.05
1.062	26.97
1.688	42.88

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. See table I for additional dimensions.

FSC 5910

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REQUIREMENTS:

Dimensions: See figure and table I.

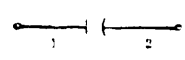
Case material: Magnetic or nonmagnetic.

Capacitance value: See table I.

Capacitance tolerance: See table I.

DC rated voltage: See table I.

Circuit diagram:



Seal:
Method 112 of MIL-STD-202, condition A.
Liquid-filled capacitors - One hour at 125°C ±3°C with no evidence of leakage.

Dielectric withstanding voltage:
Method 301 of MIL-STD-202.
Terminal to terminal - 200 percent of rated dc voltage.
Terminal to case - 400 percent of rated dc voltage or 2,000 volts, whichever is lower.

Capacitance:
Method 305 of MIL-STD-202 (for value and tolerance see table I).

Dissipation factor:
At +25°C - 1.0 percent.
At +125°C - 2.0 percent.

Insulation resistance:
Terminal to terminal - See table III.
Terminal to case (case not a terminal) - 10,000 megohms.

Barometric pressure:
Method 105 of MIL-STD-202.
Test condition - B (3.44 inches of mercury (50,000 ft)).

Vibration, low frequency:
Method 201 of MIL-STD-202.

Temperature cycling:
Method 102 of MIL-STD-202, condition D, except that for step 3, temperature shall be 125°C.

Immersion:
Method 104 of MIL-STD-202, condition C.
Insulation resistance:
Terminal to terminal - Not less than 1/3 of the value specified in table III.
Terminal to case (case not a terminal) - 5,000 megohms, minimum.
Dielectric withstanding voltage:
Terminal to terminal - 150 percent rated dc voltage.
Terminal to case (case not a terminal) - 400 percent of rated dc voltage or 2,000 volts, whichever is lower.
Capacitance change - Not to exceed 10 percent of initial value.
Dissipation factor - Not to exceed 110 percent of initial requirement.

Terminal strength:
Method 211 of MIL-STD-202, conditions A and E.
Condition A - Applied force 2 pounds.
Condition E - Torque 2 pound-inches.

Low temperature and capacitance change with temperature:
Conditions - 48 hours with rated voltage applied.
Capacitance change - See table III.

Life:
Qualification - Test voltage 140 percent of rated voltage, 1,000 hours at high ambient temperature.
Quality conformance - 250 hours at conditions specified for 1,000-hour test.
Dissipation factor:
At high ambient test temperature (any time 24 to 48 hours from start of test) - Not greater than 2.0 percent.
At high ambient test temperature (any time during last 48 hours of test) - Not greater than 2.5 percent.
At 25°C (after life test) - Not greater than 1.25 percent.

Capacitance: At 25°C (after life test) - Within 10 percent of initial value.

Insulation resistance: At 25°C (after life test):
Terminal to terminal - Not less than 1/3 of value specified in table III.
Terminal to case (case not a terminal) - Not less than 5,000 megohms.

Supersession information: See table I.

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TABLE I Dimensions and dash numbers.

Type designation 1/	DC rated voltage	Capacitance value	Dimensions, nominal (inches) F ±.062 (1.57)	T ±.062 (1.57)	W +.062 -.125 (1.57) (2.18)	G ±.031 (0.79)	L ±.062 (1.57)	Superseded MIL-C-18312/4A dash numbers. Capacitance tolerance ±10%	±20%
	Volts	µf							
CH53B1NV475-	150	4.7	1.812	1.000	0.750	2.125	2.500	0101	0102
CH53B1NV685-	150	6.8	2.000	1.750	0.750	2.375	2.750	0103	0104
CH53B1NV106-	150	10.0	2.000	1.750	0.750	2.375	2.750	0105	0106
CH53B1NV156-	150	15.0	2.000	2.000	1.125	2.375	2.750	0107	0108
CH53B1NC474-	200	0.47	1.812	1.000	0.750	2.125	2.500	0109	0110
CH53B1NC105-	200	1.0	1.812	1.000	0.750	2.125	2.500	0111	0112
CH53B1NC225-	200	2.2	1.812	1.000	0.750	2.125	2.500	0113	0114
CH53B1NE224-	400	0.22	1.812	1.000	0.750	2.125	2.500	0115	0116
CH53B1NE474-	400	0.47	1.812	1.000	0.750	2.125	2.500	0117	0118
CH53B1NE105-	400	1.0	1.812	1.000	0.875	2.125	2.500	0119	0120
CH53B1NE225-	400	2.2	2.000	1.750	0.750	2.375	2.750	0121	0122
CH53B1NE475-	400	4.7	2.000	2.000	1.000	2.375	2.750	0123	0124
CH53B1NF104-	600	0.1	1.812	1.000	0.750	2.125	2.500	0125	0126
CH53B1NF224-	600	0.22	1.812	1.000	0.750	2.125	2.500	0127	0128
CH53B1NF474-	600	0.47	1.812	1.000	0.750	2.125	2.500	0129	0130
CH53B1NF105-	600	1.0	1.812	1.250	0.875	2.125	2.500	0131	0132
CH53B1NF225-	600	2.2	2.000	2.000	0.875	2.375	2.750	0133	0134
CH53B1NF475-	600	4.7	2.000	2.000	1.500	2.375	2.750	0135	0136

1/ Complete type designation will include an additional symbol for capacitance tolerance.

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TABLE II. Millimeter equivalents of decimal inches.

Inches	MM	Inches	MM
.031	0.79	1.500	38.10
.062	1.57	1.750	44.45
.125	3.18	1.812	46.02
.750	19.05	2.000	50.80
.875	22.23	2.125	53.98
1.000	25.40	2.375	60.33
1.125	28.58	2.500	63.50
1.250	31.75	2.750	69.85

TABLE III. Characteristic.

Operating temperature range	-55 to 125 ±3°C	
DC voltage rating (volts)	150 200	400 600
High ambient test temperature	125° ±3°C	125° ±3°C
Low ambient test temperature	-55° ±3°C	-55° ±3°C
Megohms x microfarads:		
At 25°C	2000 min	2000 min
At high ambient test temperature	10 min	40 min
Insulation resistance (megohms):		
At 25°C 1/	12000	12000
At high ambient test temperature 1/	150	600
Capacitance change at low temperature	-15% max	-15% max
Life test voltage (% rated voltage)	140%	140%

1/ Need not exceed.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU
Navy - EC
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - None
Navy - AS, OS, MC, CG
Air Force - 19

Preparing activity:
Navy - EC

(Project 5910-0968-6)

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DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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DEPARTMENT OF THE NAVY
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter)

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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